

The Solar System: Worlds Around Our Sun

Planets, moons, asteroids, comets, and the environment surrounding our star

Original educational reference prepared as a compact introduction to space science.

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Chapter 1: The Sun

The Sun contains almost all of the Solar System's mass. Nuclear fusion in its core converts hydrogen into helium and releases energy that ultimately powers most surface and atmospheric processes on Earth.

Why it matters: this topic connects individual observations to the larger scientific effort to understand how celestial objects form, evolve, interact, and can be explored with modern instruments and spacecraft.

Chapter 2: Mercury

Mercury is the closest planet to the Sun and has a heavily cratered surface. With almost no substantial atmosphere, it experiences extreme temperature variations and preserves an ancient geological record.

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Chapter 3: Venus

Venus is similar to Earth in size but radically different in environment. A dense carbon-dioxide atmosphere produces an extreme greenhouse effect, while sulfuric-acid clouds cover the planet.

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Chapter 4: Earth and the Moon

Earth's liquid oceans, active geology, atmosphere, magnetic field, and biosphere make it unique among currently known worlds. The Moon stabilizes aspects of Earth's rotational behavior and preserves evidence of early Solar System impacts.

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Chapter 5: Mars

Mars is a cold desert world with volcanoes, canyons, polar ice, and extensive evidence that liquid water once flowed across its surface. It remains one of the most important targets in the search for past habitability.

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Chapter 6: Jupiter

Jupiter is the largest planet and a gas giant dominated by hydrogen and helium. Its powerful magnetic field, turbulent atmosphere, Great Red Spot, and diverse moons form a planetary system in miniature.

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Chapter 7: Saturn

Saturn is famous for its spectacular rings, composed mainly of ice particles. Its moon Titan has a thick atmosphere and hydrocarbon lakes, while Enceladus ejects water-rich material from a subsurface ocean.

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Chapter 8: Uranus and Neptune

Uranus and Neptune are ice giants containing larger proportions of water-, ammonia-, and methane-related materials than Jupiter and Saturn. Both have rings, moons, atmospheres, and unusual magnetic fields.

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Chapter 9: Asteroids and Comets

Asteroids are rocky or metallic remnants, while comets contain volatile-rich material that can form glowing comae and tails near the Sun. Both populations preserve clues about Solar System formation.

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Chapter 10: Beyond Neptune

The Kuiper Belt contains Pluto and many other icy bodies. Farther out, the hypothetical Oort Cloud is thought to contain vast numbers of cometary objects extending to the remote gravitational frontier of the Solar System.

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